

AI-Powered Governance Intelligence Platform

Document 2: Frontend Design Requirements Document (DRD / UI-UX Spec)

1. Design System & Style Guide

A. Color Tokens (Semantic & Hex)

Designed to convey institutional authority, high data density, and WCAG AA-compliant visual hierarchy.

- **Base Surfaces:**
 - `--bg-app: #0B0F19` (Ultra-deep slate navy base)
 - `--bg-surface-elevated: #111827` (Card, panel, and modal background)
 - `--bg-surface-hover: #1F2937` (Hover states for interactive surfaces)
 - `--border-subtle: #374151` (Card dividers, table borders)
 - `--border-focus: #3B82F6` (Active input outlines)
- **Brand & Interaction:**
 - `--primary-500: #2563EB` (Primary CTAs, active tab indicators)
 - `--primary-600: #1D4ED8` (Hover state for primary buttons)
 - `--primary-muted: #1E3A8A` (Selected item fills, badge backgrounds)
 - `--accent-cyan: #06B6D4` (Data visualization highlights, secondary metrics)
- **Typography & Contrast:**
 - `--text-primary: #F9FAFB` (100% white-slate for headings and primary metrics)
 - `--text-secondary: #9CA3AF` (60% gray for table labels, subtitles)
 - `--text-tertiary: #6B7280` (Disabled text, placeholder text)
- **Alert & Anomaly States:**
 - `--alert-critical-text: #FCA5A5` | `--alert-critical-bg: #7F1D1D` (Low utilization <40%, stalled projects)
 - `--alert-warning-text: #FDE68A` | `--alert-warning-bg: #78350F` (Divergence/Overlaps, reporting delays)
 - `--alert-success-text: #86EFAC` | `--alert-success-bg: #14532D` (Target achieved, verified convergence)

B. Typography Scale

Font Family: **Inter** (Primary UI) with **JetBrains Mono** for numbers, LGD codes, and financial amounts.

Level	Font Family	Size (px)	Line Height (px)	Weight	Tracking	Usage
Display 1	Inter ▾	32	40	Bold (700) ▾	-0.02em	Executive KPI Headline
Heading 1	Inter ▾	24	32	Semi-Bold (600) ▾	-0.01em	View/Dashboard Titles
Heading 2	Inter ▾	18	26	Semi-Bold (600) ▾	Normal	Card Headers, Modal Titles
Body Regular	Inter ▾	14	20	Regular (400) ▾	Normal	General Descriptions, Form Labels
Body Medium	Inter ▾	14	20	Medium (500) ▾	Normal	Table Rows, Button Labels
Metric Value	JetBrains Mono ▾	20	28	Bold (700) ▾	-0.01em	Financial Figures (₹ Cr), Counts
Caption / Code	JetBrains Mono ▾	12	16	Regular (400) ▾	+0.02em	LGD Codes, Timestamps, Citations

C. Core Component Interactive States

- Primary Button:
 - Default: Background --primary-500, Text --text-primary, Border none, Radius 6px, Height 38px, Padding 0 16px.
 - Hover: Background --primary-600, Transition background-color 150ms ease-in-out.
 - Active: Background #1E40AF, Scale 0.98.
 - Disabled: Background --bg-surface-hover, Text --text-tertiary, Cursor not-allowed, Opacity 0.5.
- Search / NL Input Box:
 - Default: Background --bg-surface-elevated, Border 1px solid --border-subtle, Radius 8px, Text --text-primary, Height 48px, Left Icon (Search), Right Icon (Command+K or Mic).
 - Hover: Border 1px solid #4B5563.
 - Active/Focus: Border 1.5px solid --primary-500, Box-shadow 0 0 0 3px rgba(59, 130, 246, 0.25), Outline none.
 - Disabled: Background --bg-app, Border 1px solid --border-subtle, Cursor not-allowed.

2. Layout & Responsive Breakpoints

- **Breakpoint Standards:**
 - Desktop (Default): $\geq 1280\text{px}$ (Optimized for 1440×900 and 1920×1080 command centers)
 - Tablet (Landscape): $1024\text{px} - 1279\text{px}$
 - Tablet (Portrait): $768\text{px} - 1023\text{px}$
 - Mobile: $< 768\text{px}$
- **Desktop Layout Container:**
 - Persistent left sidebar (260px width), fluid main workspace, max inner content container 1600px centered with 24px/32px padding.
 - Flexbox row for filter bars; 12-column CSS Grid with 20px gap for metric cards.
- **Tablet Layout Container:**
 - Sidebar collapses to an icon-only 72px rail.
 - Metric cards collapse from 4-columns to 2-columns ($\text{repeat}(6, 1\text{fr})$).
 - Wide data tables activate horizontal scrolling with the first two columns (Scheme Name, LGD Entity) sticky pinned.
- **Mobile Layout Container:**
 - Sidebar converts to an off-canvas drawer toggled via 56px hamburger header.
 - Grid transforms to single-column vertical stack (100% width).
 - Interactive choropleth map features a touch zoom lock with an explicit "Use two fingers to pan" overlay.

3. Visual State Matrix

State Type	Visual Implementation & CSS Rules	Behavior / Content
Empty State	Centered flexbox, padding 64px 24px. Border: dashed 1px solid <code>--border-subtle</code> . Vector illustration (monochrome slate).	Feedback: "No anomaly records match the selected filters for District: Basti." Primary CTA: "Reset Filters".
Skeleton Loading	Placeholder geometric blocks matching target dimensions. Gradient shimmer 200% width, animation 1.5s infinite.	Prevents Cumulative Layout Shift (CLS). Tables render 5 rows of skeleton lines; KPI cards render skeleton metrics.
Error Banner	Fixed top banner or inline card alert. Background <code>--alert-critical-bg</code> , border 1px solid	Text: "Data sync failure: Unable to fetch LGD boundary polygon from NIC GIS service (HTTP 504)." Action: "Retry Sync".

State Type	Visual Implementation & CSS Rules	Behavior / Content
	<code>--alert-critical-text</code> . Icon: Alert Triangle.	
Toast Alert	Floating bottom-right (bottom: 24px, right: 24px, z-index: 9999). Width 360px, slide-in-up 200ms.	Displays asynchronous status updates (e.g., "Natural language query exported to CSV successfully"). Auto-dismiss 4000ms.

4. Accessibility & Usability Standards

- **Contrast Compliance:** All text-to-background contrast ratios strictly meet WCAG 2.1 Level AA standards: minimum 4.5:1 for normal body text and 3:1 for graphical UI components and large text.
- **Touch Targets:** All interactive elements provide a minimum bounding box of 44 x 44 pixels on touch-enabled tablet and mobile interfaces.
- **Keyboard Navigation:** Full Tab Order navigation enabled. Every interactive node contains an explicit focus ring (`focus-visible: 2px solid --primary-500; outline-offset: 2px`). Map pan/zoom supports standard arrow keys (+, -, and arrow keys).
- **Screen Reader Semantic Roles:**
 - Dynamic live metrics marked with `aria-live="polite"`.
 - Anomaly banners flagged with `role="alert"`.
 - Expandable evidence drawers linked with `aria-expanded="true/false"` and `aria-controls="evidence-panel-id"`.